Journal of Multilingual and Multicultural Development

ISSN: 0143-4632 (Print) 1747-7557 (Online) Journal homepage: www.tandfonline.com/journals/rmmm20

# Distinct language control mechanisms in speech production and comprehension: evidence from *N*-2 repetition, switching, and mixing costs

Junjie Wu, Chuyao Cai, Minzhe Wu, Xin Wang & Miao Yu

**To cite this article:** Junjie Wu, Chuyao Cai, Minzhe Wu, Xin Wang & Miao Yu (27 Jun 2025): Distinct language control mechanisms in speech production and comprehension: evidence from *N*-2 repetition, switching, and mixing costs, Journal of Multilingual and Multicultural Development, DOI: 10.1080/01434632.2025.2522829

**To link to this article:** https://doi.org/10.1080/01434632.2025.2522829

JOURNAL OF MULTILINGUAL AND MULTICULTURAL DEVELOPMENT
https://doi.org/10.1080/01434632.2025.2522829



# Distinct language control mechanisms in speech production and comprehension: evidence from *N*-2 repetition, switching, and mixing costs

Junjie Wu[a,b,c,d*], Chuyao Cai[a,c*], Minzhe Wu[a,c], Xin Wang[b] and Miao Yu[a,c,d]

[a]Key Research Base of Humanities and Social Sciences of the Ministry of Education, Academy of Psychology and Behavior, Tianjin Normal University, Tianjin, People's Republic of China; [b]Department of Linguistics, Faculty of Medicine, Health and Human Sciences, Macquarie University, Sydney, Australia; [c]Faculty of Psychology, Tianjin Normal University, Tianjin, People's Republic of China; [d]Tianjin Key Laboratory of Student Mental Health and Intelligence Assessment, Tianjin, People's Republic of China

**ABSTRACT**

The debate remains unresolved regarding whether identical language control mechanisms underlie production and comprehension in individuals who use two or more languages. Most previous research focused solely on switching costs aggregated at the group level, potentially overlooking a broader set of indices and individual variabilities. To address this, we integrated the *n*-2 repetition paradigm, language-switching paradigm, and single-language tasks to examine two modalities in Cantonese–Mandarin–English trilinguals: spoken word production (Experiment 1) and auditory word comprehension (Experiment 2). We evaluated a wider range of language control indices, including *n*-2 repetition, switching, and mixing costs. Group-level analyses revealed significant effects for all indices in production, whereas only mixing costs reached significance in comprehension. Individual differences analyses showed no significant correlations between production and comprehension across three types of costs, with Bayesian statistics providing supporting evidence for the absence of such relationships in most cases. Radar and permutation analyses further showed that language control in production was quantitatively greater than comprehension, and qualitatively distinct in pattern from comprehension. These findings show that language control in production is not only stronger, but also qualitatively distinct from that in comprehension, underscoring the importance of distinguishing between modalities in models of language control.

**ARTICLE HISTORY**
Received 9 January 2025
Accepted 13 June 2025

**KEYWORDS**
Language switching; multilingualism; bilingualism; Cantonese–Mandarin–English trilinguals; modality-specific mechanisms; shared mechanisms

## Research highlights

(1) The debate persists on shared language control in production and comprehension.
(2) We used comprehensive indices: *n*-2 repetition, switching, and mixing costs.
(3) All indices were significant in production, but only mixing cost in comprehension.
(4) The two modalities were uncorrelated and differed in both magnitude and pattern.
(5) Language control may differ qualitatively between production and comprehension.

**CONTACT** Miao Yu ym@tjnu.edu.cn
*Shared first-authorship
Supplemental data for this article can be accessed online at https://doi.org/10.1080/01434632.2025.2522829.

## Introduction

For individuals who speak two or more languages (i.e. bilinguals or multilinguals), both target and non-target languages are activated during language production (e.g. Costa, Miozzo, and Caramazza 1999; Schwieter and Sunderman 2011) and comprehension (e.g. Lim and Ahn 2024; Thierry and Wu 2007), causing cross-language competition that necessitates language control mechanisms (e.g. Abutalebi and Green 2008; Hermans et al. 1998). The question of whether these mechanisms are identical or different in language production and comprehension remains a significant debate. Most previous studies have approached this issue by using switching cost as an index, aggregating it at the group level, and evaluating its existence across production and comprehension (e.g. de Bruin and Xu 2023; Gambi and Hartsuiker 2016). However, this approach may limit insights due to its reliance on a single index, which lacks a comprehensive array of indices and tends to overlook individual variabilities. The present study addressed this issue by measuring the *n*-2 repetition, switching, and mixing cost as indices of language control among Cantonese–Mandarin–English trilinguals engaged in language-switching tasks during spoken word production and auditory word comprehension.

The language-switching task has been extensively employed to investigate language control mechanisms (e.g. Meuter and Allport 1999; Wu et al. 2019). This task, in production studies, requires participants to name pictures in one of their languages based on an unpredictable cue, resulting in either repeat or switch trials. Trials that use the same language as the previous trial are referred to as repeat trials (e.g. L1 → **L1**), and trials that use a different language from the preceding trial are termed switch trials (e.g. L1 → **L2**). Participants typically exhibit longer reaction times (RTs) and higher error rates (ERs) on switch trials compared to repeat trials, a phenomenon known as switching costs (switching costs = switch – repeat). Switching costs reflect the effort required to resolve trial-by-trial language conflicts incurred by the activation of the non-target language, and serve as an index of local control (Declerck and Koch 2022; Wang et al. 2022). Similarly, comprehension studies use language-switching tasks where language is cued by the input itself (e.g. Declerck et al. 2019; Declerck and Grainger 2017; Wu et al. 2025). In these tasks, participants judge whether visually or auditorily presented words from two languages refer to living or non-living objects.

In production, language control is commonly examined within the theoretical framework of the Inhibitory Control Model (ICM, Green 1998). According to this model, when individuals are required to choose a specific language for production, lexical items in both languages are non-selectively activated. A language tag is then employed to classify the words into their respective language categories. Subsequently, the language task schema selects the target language while suppressing the non-target language at the lemma level. Importantly, substantial empirical evidence from the language-switching task, e.g. switching costs, supports the ICM and confirms the role of language control mechanisms in production (e.g. Green 1998; Ma, Li, and Guo 2016; Meuter and Allport 1999; Wang et al. 2022; Yang et al. 2024).

In comprehension, language control is typically examined within the theoretical framework of the Bilingual Interactive Activation Model (BIA, Dijkstra and van Heuven 1998) and Bilingual Interactive Activation Plus Model (BIA+, Dijkstra and van Heuven 2002). The BIA model describes bilingual word recognition as an automatic, bottom-up process where visual input activates words from both languages. Competition among words, which share similar orthography or phonology, is resolved through mutual inhibition and feedback from language nodes, which suppress non-target language words. However, the BIA model confines control to the lexical level, is insensitive to task demands or context, and attributes language switching to lexical-level processes only. Therefore, Dijkstra and van Heuven (2002) extended BIA to BIA + by redefining language nodes as passive tags and introducing a separate task/decision system. This shifts language control and switching to a higher, post-lexical level, where task goals and contextual factors guide selection. This key extension allows BIA + to account for flexible, context-sensitive language control essential for managing cross-language interference in both visual and auditory comprehension, and potentially in production.

Moreover, one proposal holds that language control in production and comprehension is governed by shared underlying mechanisms, with differences arising only in their mode of triggering – exogenous in comprehension and endogenous in production (Grainger, Midgley, and Holcomb 2010). However, in contrast to production (see Gade et al. 2021, for a meta-analysis), switching costs were not always observed in comprehension studies (Declerck et al. 2019; Declerck and Grainger 2017; but see Hirsch, Declerck, and Koch 2015; Thomas and Allport 2000; Wu et al. 2025), thereby raising doubts about the notion that identical control mechanisms operate in both language production and comprehension.

Several studies have investigated this issue by examining switching costs in both language production and comprehension within the same study (e.g. Ahn et al. 2020; Macizo, Bajo, and Paolieri 2012; Reynolds, Schlöffel, and Peressotti 2016), and even within a single task (e.g. Gambi and Hartsuiker 2016; Peeters et al. 2014). For example, Gambi and Hartsuiker (2016) instructed pairs of Dutch-English bilinguals to take turns naming pictures, with one participant naming in one trial and the other in the next. Participants exhibited longer RTs when naming in L1 after hearing their partners name in L2, compared to after L1. They interpreted this as evidence that both production and comprehension rely on the same language control system. However, this approach may introduce confounds related to task switching (Li and Gollan 2022) or shared language processing rather than language control (Eichert et al. 2020; Liang et al. 2023).

Therefore, other studies, which employed separate tasks for production and comprehension within the same study, avoided these potential confounds, but they primarily relied on switching costs that are fairly limited in the scope of their investigation into language control (Ahn et al. 2020; Blanco-Elorrieta and Pylkkanen 2016, 2017; Li et al. 2024; Macizo, Bajo, and Paolieri 2012; Qi et al. 2009; Qi, Peng, and Ding 2010; Reynolds, Schlöffel, and Peressotti 2016). For instance, Qi et al. (2009) found different patterns of switching costs: greater switching costs for L1 in production, but for L2 in comprehension. Similarly, Li et al. (2024) also found different patterns using ERP measures for comprehension and speech errors for production: switching to the nondominant language elicited an early, long-lasting positive potential in comprehension, whereas switching to the dominant language resulted in more intrusion errors in production. Although these studies did not conduct direct statistical comparisons between modalities, they argued for distinct language control mechanisms in production and comprehension based on different patterns aggregated at the group level, an approach that may overlook individual differences (Miller and Schwarz 2018).

Moreover, some studies have gained insights from the perspective of individual differences, specifically by examining how variability in one modality affects performance in the other (Ahn et al. 2020; Struys et al. 2019). The underlying rationale is that if two tasks/modalities engage (partially) overlapping cognitive mechanisms, then individual differences in the efficiency of engaging them should result in correlated performance across tasks. Specifically, Struys et al. (2019) found a few significant correlations between verbal fluency production and language-switching comprehension tasks, proposing shared language control processes. Ahn et al. (2020) reported little correlation between intrusion errors in production and switching costs measured by eye-tracking indices in comprehension, favouring separate control mechanisms cross-modalities. However, the intrusion errors and verbal fluency measures used in these studies reflect both language processing and control. Measures specifically targeting language control primarily relied on switching costs alone, which may constrain a more comprehensive understanding of language control.

The *n*-2 language repetition cost is regarded as a purer index of inhibitory control compared to the switching cost (Philipp, Gade, and Koch 2007; Philipp and Koch 2009), despite the latter being a classic measure in the literature. Specifically, the switching cost may arise from two possible sources: resisting persistent activation of the non-target language or overcoming persistent inhibition of the target language from the previous trial. To address this issue, *n*-2 repetition cost was developed through the $n$-2 paradigm, in which trilingual participants switch unpredictably among three languages. Within this paradigm, ABA sequences feature the same language in trials $n$ and $n$-2 (e.g. L1–L3–L1). Participants return to and repeat Language A in trial $n$, which had been used in trial $n$-2 but inhibited in trial $n$-1. In contrast, CBA sequences involve different languages in each trial (e.g. L2–L3–L1). Participants use Language A in trial $n$, which had not been used in trials $n$-2 and $n$-1. If persistent inhibition of Language A drives the process, trilinguals will exhibit longer RTs and higher ERs in ABA sequences than in CBA sequences, which is the $n$-2 repetition cost. Conversely, persistent activation of Language A would yield the opposite pattern, with shorter RTs and lower ERs in ABA than in CBA sequences (i.e. $n$-2 repetition benefit). In fact, numerous studies have consistently revealed robust $n$-2 repetition costs in language production (e.g. de Bruin, Hoversten, and Martin 2023; Declerck and Philipp 2017; Koch et al. 2024; Philipp, Gade, and Koch 2007). These findings support the role of persistent inhibition in language control, and show the $n$-2 repetition cost as a more solid index of inhibitory, compared to switching costs, though both indexing local control through resolving conflict on a trial-by-trial basis. Moreover, Declerck and Philipp (2017) found that production tasks exhibited $n$-2 repetition costs, whereas the comprehension task did not. They suggested that both production and comprehension engage the same inhibitory mechanisms via language nodes, but production involves a quantitatively greater activation of the corresponding language node. Notably, while $n$-2 repetition costs and switching costs are robust in production studies when collapsed across languages, closer examination of language-specific costs reveals inconsistent findings, challenging the ICM assumption that the dominant language is more strongly inhibited and thus should show larger costs (Koch et al. 2024; see Declerck and Koch 2022 for a review). One possibility is that switching costs and $n$-2 repetition costs, both reflecting local control, may be modulated by global control. Specifically, global control may over-inhibit the dominant languages, rendering them contextually non-dominant and resulting in local control indices that do not reliably align with language dominance. However, this account warrants further investigation using indices that capture language control across multiple levels.

In addition, the mixing cost is widely regarded as an index of global control (e.g. Braver 2012; Declerck and Koch 2022; Ma, Li, and Guo 2016). It refers to the additional RTs and ERs observed in repeat trials of language-switching tasks (also known as mixed-language contexts) compared to trials in single-language contexts (mixing costs = repeat − single), reflecting the global and proactive inhibition applied to the non-target language throughout the language-switching task (Braver 2012; Yang et al. 2024). Mixing cost is robust in production (e.g. Wu et al. 2020, 2025), whereas it is less reliably detectable in comprehension (Declerck et al. 2019). Specifically, in the study of Declerck et al. (2019), mixing costs were observed in only one of the three experiments involving the processing of written words.

In summary, ongoing debate surrounds whether the language control mechanisms underlying language production and comprehension are identical (Gambi and Hartsuiker 2016; Peeters et al. 2014) or differ quantitatively (de Bruin and Xu 2023; Declerck and Philipp 2017; Macizo, Bajo, and Paolieri 2012) or qualitatively (e.g. Ahn et al. 2020; Blanco-Elorrieta and Pylkkanen 2016; Li et al. 2024; Mosca and de Bot 2017). In the present study, we aimed to address this issue by using a comprehensive set of indices while incorporating an individual difference approach. Specifically, we recruited Cantonese–Mandarin–English trilinguals to perform spoken word production (Experiment 1) and/or auditory word comprehension (Experiment 2) in both mixed- and single-language contexts. We hypothesise that if the same set of underlying mechanisms operates across production and comprehension – whether engaged to the same degree (i.e. identical) or to varying degrees (i.e. quantitative difference) – then $n$-2 repetition, switching, and mixing costs would be similar or differ in magnitude, respectively, while remaining correlated and exhibiting comparable patterns across modalities. In contrast, if distinct sets of mechanisms are engaged across modalities (i.e. qualitative difference), these language control indices would differ in pattern and exhibit nonsignificant correlations. More specifically, the magnitudes of language control indices across modalities will be evaluated through polygon area and permutation tests, while patterns will be assessed using cosine similarity alongside permutation tests.

## Experiment 1

Experiment 1 aimed to investigate the language control mechanisms in spoken word production of Cantonese–Mandarin–English trilinguals by measuring *n*-2 repetition costs, switching costs, and mixing costs.

### *Participants*

To determine the sample size for our study, we conducted a power analysis using G*Power (Faul et al. 2007). We conducted a two-tailed paired *t*-test with an alpha level of 0.05 and a medium effect size ($d$) of 0.5, determining that a sample size of 34 participants would achieve a statistical power of 0.80. To account for potential attrition or unforeseen issues, we recruited more participants than required.

Forty-one trilinguals (18 males, age = 20.95 ± 2.21 [$M \pm SD$]) were recruited for a production task. Participants' language history questionnaires showed that Cantonese, Mandarin, and English were acquired as L1 (2.93 ± 4.46 years), L2 (4.35 ± 3.69 years), and L3 (7.75 ± 2.31 years), respectively. Self-rated proficiency was much higher for L1 and L2 compared to L3 (7.97 ± 1.21, 8.30 ± 0.88, and 6.06 ± 1.40; $t$s > 7.86, $p$s < 0.001), with L2 slightly higher L1 ($t = 2.16$, $p = 0.037$). Over the past year, participants predominantly used Mandarin (63.83% ± 19.03%), followed by Cantonese (21.54% ± 16.84%) and English (12.85% ± 9.52%). Refer to the *Supplementary Material* for detailed information on participants' language backgrounds. All participants had normal or corrected-to-normal vision and self-reported normal hearing. Written informed consent was obtained prior to the experiment and all participants were compensated for their time.

### *Materials and design*

Twenty-six standardized black-and-white line drawings were selected from Snodgrass and Vanderwart (1980), with 20 used for the formal experiment and 6 used as practice. See the *Supplementary Material* for detailed information on material characteristics.

This study employed a within-subject design incorporating two independent variables: Language (L1, L2, and L3) and Trial type. Trial type varied depending on the specific indices: *n*-2 repetition costs (ABA vs. CBA), switching costs (switch vs. repeat), and mixing costs (repeat vs. single). This design was applied to both production and comprehension tasks.

The production task consisted of both single- and mixed-language contexts. In the single-language contexts, participants named pictures in only one language per block. Three blocks for the single-language contexts included 180 trials in total, equally split among Cantonese, Mandarin, and English, with 20 formal pictures in each block repeated an equal number of times. In the mixed-language contexts, participants named pictures in Cantonese, Mandarin or English according to unpredictable cues (i.e. a red, green, or blue square) across 480 trials evenly divided among the three languages, with each using 20 formal pictures repeated equally. Trials were pseudorandomized, including 232 ABA trials (76, 77, and 79 trials for L1, L2, and L3, respectively) and 71 CBA trials (25, 22, and 24 trials, respectively) for *n*-2 repetition costs, as well as 391 switch trials (130, 130, and 131 trials, respectively) and 88 repeat trials (30, 29, and 29 trials, respectively) for switching costs. Notably, the first trial of each switching cost sequence and the first two trials of each *n*-2 repetition cost sequence were treated as fillers and were not classified as critical trials. Our trial lists combined the language-switching and *n*-2 repetition paradigms, allowing certain trials to be classified under both frameworks simultaneously; see the *Supplementary Material* for implementation details.

### Procedure

The experiment was programmed using E-prime 3.0. Participants received instructions to name pictures quickly and accurately. After practicing and learning the picture names, each trial began with a 350 ms fixation, followed by a 150 ms blank. A picture and language cue were presented simultaneously and remained on the screen until a vocal response was given. The interval between the response of the participant and the next trial was 500 ms (Figure 1(A)). Vocal responses were recorded via a microphone, and naming latencies were measured automatically by Chronos. Errors were recorded manually by the experimenter.

### Data analysis

We first pre-process the data (e.g. exclusion of extreme values and error trials) and then analyse RT and ER using mixed effects models. Language was dummy-coded, with L1, L2, or L3 as baseline. When L1 was the baseline, it was assigned [0 0], L2 [1 0], and L3 [0 1]; when L2 was the baseline, L1 was coded as [0 1], L2 as [0 0], and L3 as [1 0]; and when L3 was the baseline, L1 was coded as [1

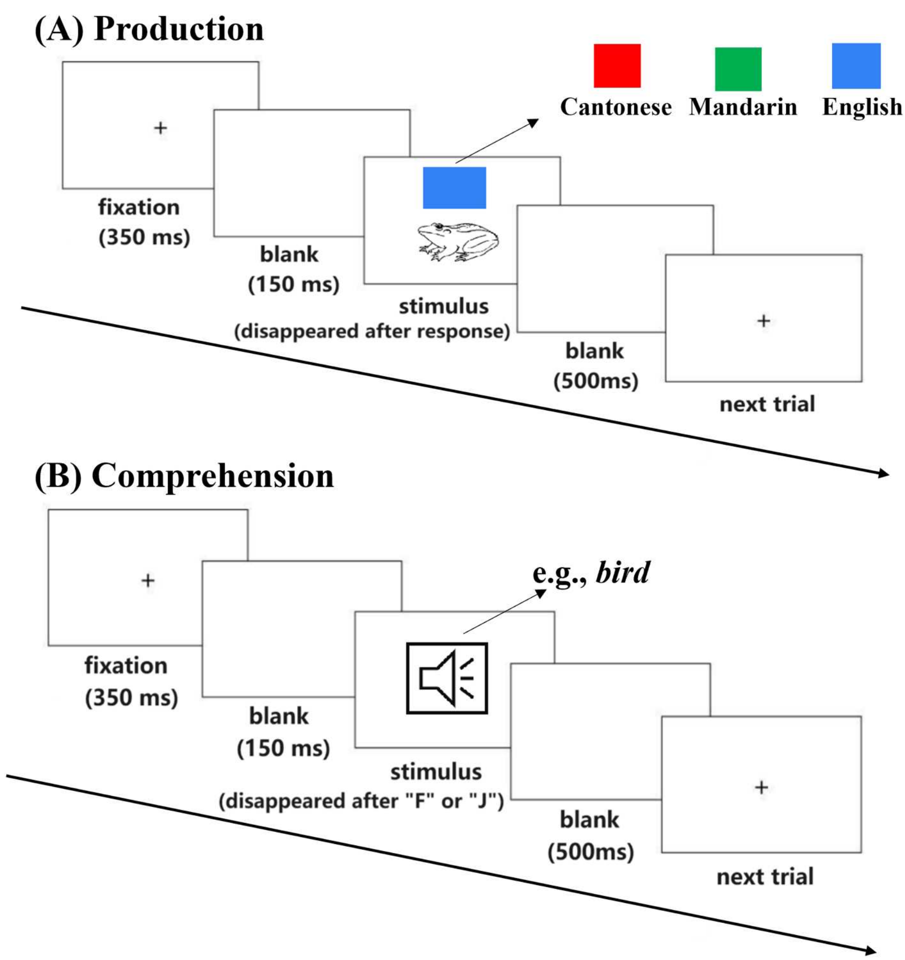


**Figure 1.** Overview of experimental paradigm. In Experiment 1 (A), participants named pictures in the appropriate language based on the square's colour: red for Cantonese, green for Mandarin, and blue for English. In Experiment 2 (B), participants listened to recordings presented in a pseudorandom order in Cantonese, Mandarin, or English. They were asked to judge whether the word referred to was a living object or not. F is correct, while J is incorrect.

0], L2 as [0 1], and L3 as [0 0]. Fixed effects included Language (L1, L2, L3), Trial type (for *n*-2 repetition costs: ABA, CBA; for switching costs: switch, repeat; for mixing costs: repeat, single), and their interactions. Random intercepts were specified for participants and items, along with random slopes for the fixed effects. The primary focus was on the main effect of Trial type to assess the presence of the costs collapsing across three languages. Additionally, LMEMs were refitted separately for each language pair (L1 vs L2, L1 vs L3, L2 vs L3) to examine specific interaction patterns, such as differences in *n*-2 repetition costs between L1 and L2. Further methodological details are provided in the *Supplementary Material*.

### Results

For the overall RT effect combining all three languages, we found significant *n*-2 repetition costs (76 ms; $F = 5.34$, $p = 0.036$), switching costs (434 ms; $F = 297.04$, $p < 0.001$), and mixing costs (331 ms; $F = 570.64$, $p < 0.001$). These costs for each language are presented in Figure 2 and Table 1. No opposite effects were observed in ER analysis, indicating the absence of a speed–accuracy trade-off. Detailed results are provided in the *Supplementary Material*.

## Experiment 2

Experiment 2 aimed to investigate the language control mechanisms in auditory word comprehension among Cantonese–Mandarin–English trilinguals.

### Participants

Forty-one students (17 males, age = 20.95 ± 2.24) participated in comprehension task, and 37 of them also participated in production task. The language background of participants was similar to that of Experiment 1 (see *Supplementary Material* for details). Cantonese, Mandarin, and English served as L1 (2.43 ± 3.42 years), L2 (4.56 ± 3.23), and L3 (7.29 ± 2.08), respectively. Self-rated proficiency was much higher for L1 and L2 than for L3 (8.00 ± 1.24, 8.34 ± 0.91, 6.10 ± 1.32; $t$s > 7.58; $p$s < 0.001), and L2 was rated slightly higher than L1 ($t = 2.16$, $p = 0.037$).

### Materials and design

Twenty-four words were selected, with 20 used in the formal experiment and 4 used as practice. All words were recorded in standard Cantonese, Mandarin, and English, with audio durations standardised to 500 ms. The materials in Experiment 2 were not entirely identical to those in Experiment 1, as some items from the production task were not fully suitable for comprehension. No significant differences between two sets across psycholinguistic variables ($p$s > 0.104; see *Supplementary Material* for details).

The design for Experiment 2 was identical to that for Experiment 1, except for minor differences in trial distributions within mixed-language contexts. For *n*-2 repetition costs, there were 228 ABA trials (76, 75, and 77 trials for L1, L2, and L3, respectively) and 69 CBA trials (23, 22, and 24, respectively). For switching costs, there were 388 switch trials (130, 129, and 129, respectively) and 91 repeat trials (30, 31, and 30, respectively).

### Procedure

The procedure (Figure 1(B)) was similar to that of Experiment 1 except participants were instructed to judge whether the word they heard referred to a living object or not by pressing either ‘F’ or ‘J’ on the keyboard. The key assignment was counterbalanced across participants.

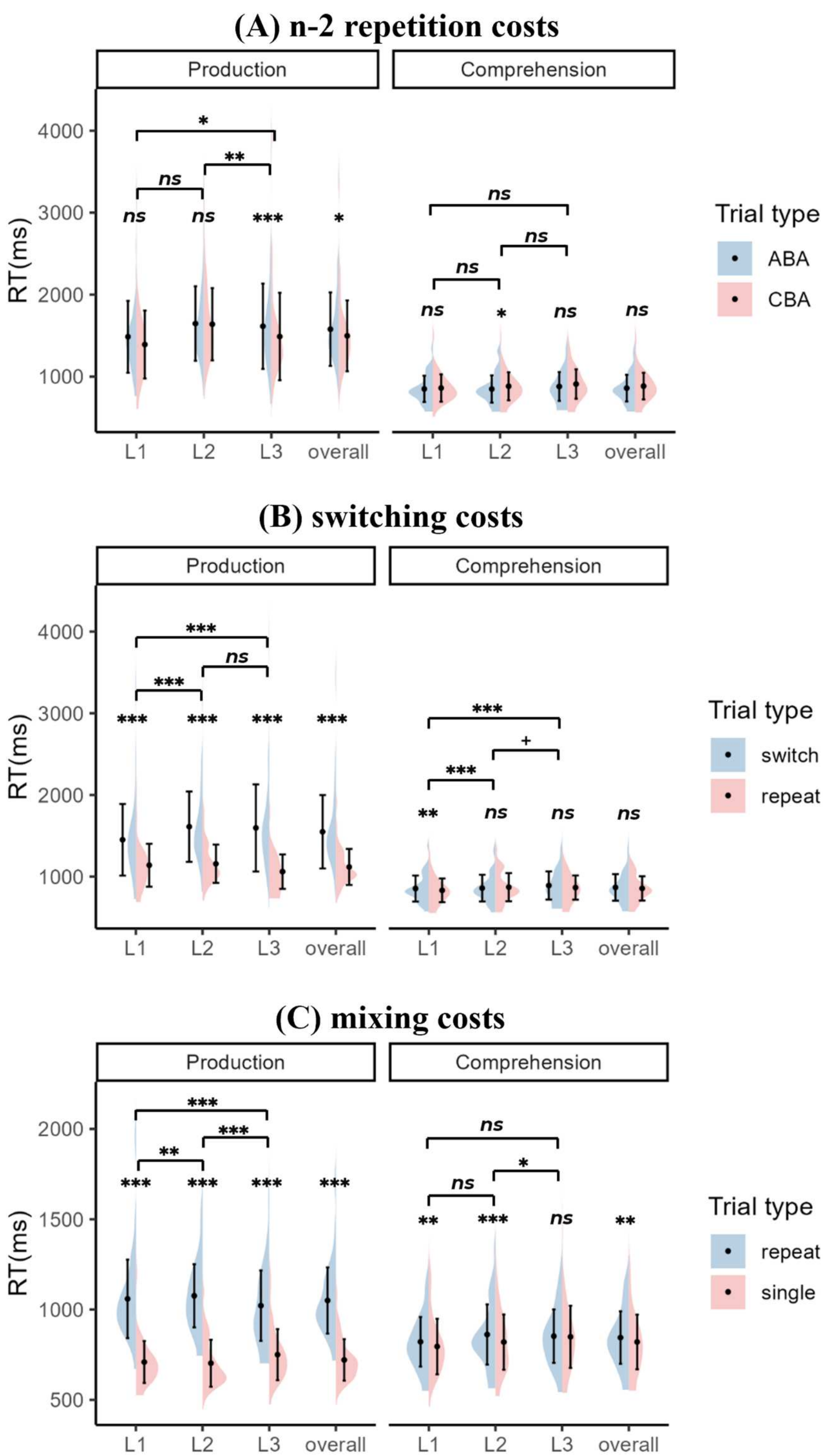


**Figure 2.** Mean *n*-2 repetition costs, switching costs and mixing costs for RT in production and comprehension. The asterisks, plus signs, and '*ns*' represent significance of differences between Trial types (e.g. switch vs. repeat; i.e. costs) as well as differences in costs between languages. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; +, $p < 0.1$; *ns*, $p \geq 0.1$. The black dots correspond to the mean, and the error bars represent one standard deviation.

**Table 1.** Overview of the effects of language on three types of costs.

| Modalities | Indices | Overall | L1 | L2 | L3 | L1 vs L2 | L1 vs L3 | L2 vs L3 | L2 ≥ L1 > L3 | Counts |
|---|---|---|---|---|---|---|---|---|---|---|
| Production | *n*-2 repetition costs | 76* | 95[ns] | 8[ns] | 126*** | L1 = L2 | L1 < L3 | L2 < L3 | × | 2 |
| | switching costs | 434*** | 311*** | 454*** | 536*** | L1 < L2 | L1 < L3 | L2 = L3 | × | 2 |
| | mixing costs | 331*** | 349*** | 373*** | 270*** | L1 < L2 | L1 > L3 | L2 > L3 | √ | 16 |
| Comprehension | *n*-2 repetition costs | −25[ns] | −10[ns] | −34* | −30[ns] | L1 = L2 | L1 = L3 | L2 = L3 | = | 4 |
| | switching costs | 12[ns] | 23** | −12[ns] | 26[ns] | L1 > L2 | L1 > L3 | L2 = L3 | × | 3 |
| | mixing costs | 24** | 26** | 42*** | 3[ns] | L1 = L2 | L1 = L3 | L2 > L3 | √ | 8[a] |

*Note.* L1, L2, and L3 refer to participants' first (Cantonese), second (Mandarin), and third (English) languages, with Mandarin rated slightly higher than Cantonese despite both being native languages, and English rated substantially lower than either. The check (√), cross (×), and equal (=) symbols indicate whether the more dominant language(s) exhibited larger costs (i.e. L2 ≥ L1 > L3), an opposite pattern, or no difference among the three languages at the group level, respectively. Counts represent the number of participants with the pattern L2 ≥ L1 > L3 among the 37 participants shared between Experiment 1 (production) and Experiment 2 (comprehension). Superscript 'a' indicates that four participants exhibited the L2 ≥ L1 > L3 pattern for mixing costs in both production and comprehension. The asterisks and '*ns*' indicate the significance levels of each cost relative to zero: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; *ns*, $p \geq 0.1$.

### *Data analysis*

Data processing procedures were identical to those used in Experiment 1, with details again provided in the *Supplementary Material.*

### *Results*

Among the three cost types (collapsed across language), only mixing costs were significant (24 ms; $F = 10.16$, $p = 0.003$), whereas *n*-2 repetition (−25 ms; $F = 2.20$, $p = 0.159$) and switching costs (12 ms; $F = 0.14$, $p = 0.709$) were not. These costs for each language are presented in Figure 2 and Table 1. No opposite effects were observed in ER analysis, indicating the absence of a speed–accuracy trade-off. Detailed results are provided in the *Supplementary Material.*

## Combined analysis

Since the previous section has conducted group-level analyses following prior studies, this section integrates data from production (Experiment 1) and comprehension (Experiment 2) from the perspective of individual differences, aiming to gain insights into whether identical language control mechanisms are involved in both modalities.

Specifically, we first conducted a Pearson correlation analysis between the two modalities for the *n*-2 repetition, switching, and mixing costs, using data from 37 participants who completed both experiments. Alongside null hypothesis significance testing (NHST), Bayesian statistics were employed to calculate *Bayes factors* (*BF*) using JASP (JASP Team 2018). $BF_{01}$ and $BF_{10}$, which are reciprocals of each other, quantified the strength of evidence for the null hypothesis over the alternative or vice versa, respectively. A Bayes Factor (*BF*) close to 1 indicates negligible evidence for either hypothesis. A *BF* greater than 3 is generally regarded as providing substantial support for the favoured hypothesis, while a *BF* exceeding 5 is considered strong evidence (Jarosz and Wiley 2014). As shown in Figure 3, we found nonsignificant correlations and substantial evidence supporting the absence of correlations between production and comprehension for overall *n*-2 repetition costs, overall switching costs, and overall mixing costs ($|r|\text{s} < 0.10$, $ps > 0.544$, $BF_{01}\text{s} > 4.08$). Note that we remove an outlier when fitting the overall index of switching costs (Figure 3(B)). When examining specific languages, these three types of costs also showed nonsignificant correlations in L1 ($|r|\text{s} < 0.31$, $ps > 0.061$) and L2 ($|r|\text{s} < 0.25$, $ps > 0.137$) and L3 ($|r|\text{s} < 0.20$, $ps > 0.238$). Furthermore, there was substantial evidence supporting the absence of correlations in the *n*-2 repetition costs across the three languages ($BF_{01}\text{s} > 3.36$), as well as in L3 mixing costs ($BF_{01} = 4.85$).

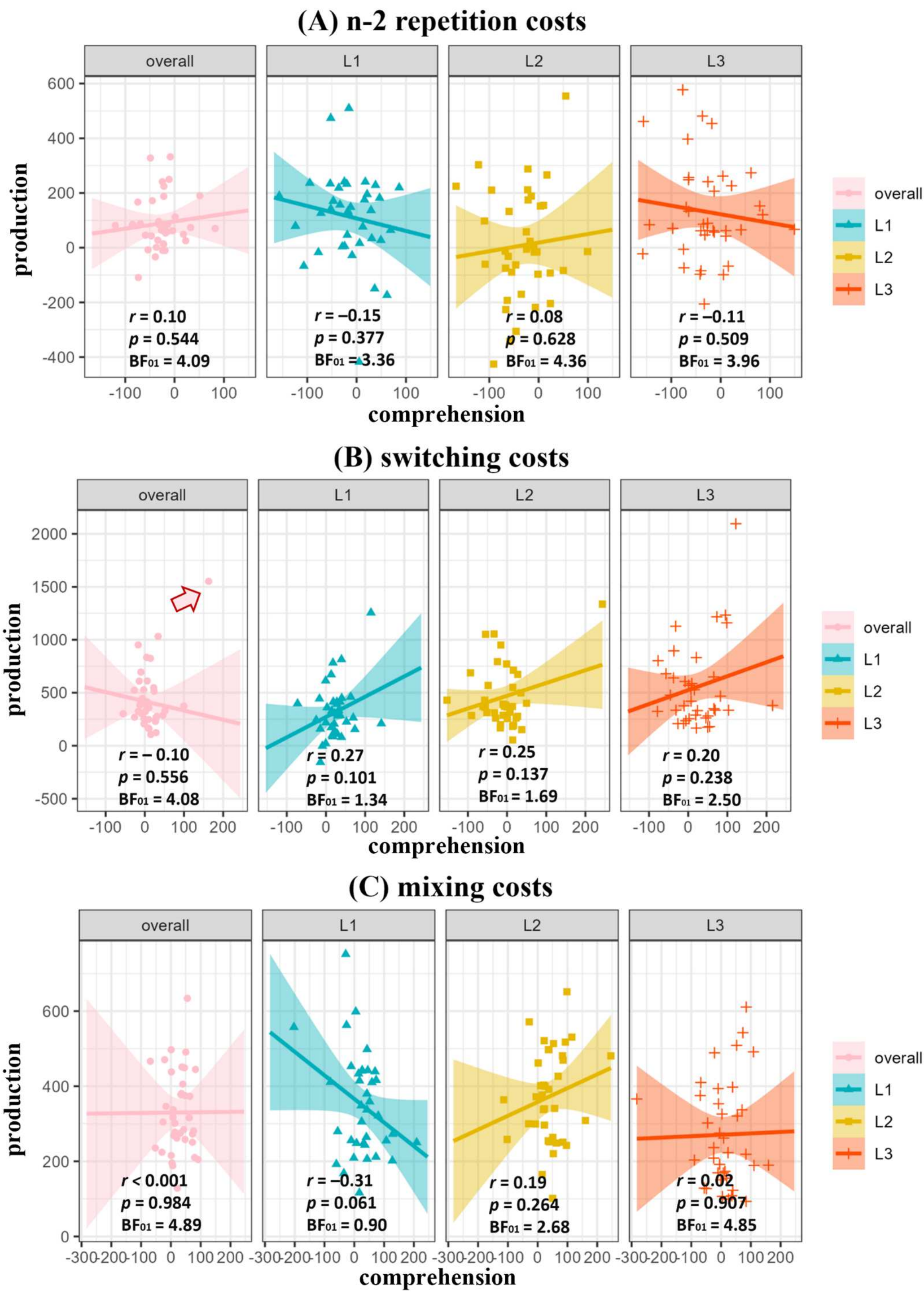


**Figure 3.** Correlations between production and comprehension for *n*-2 repetition, switching, and mixing costs. Overall refers to the combined performance across all three languages. The lines represent the fitted linear regression lines, and the shaded bands represent the 95% confidence interval. Notably, the arrow indicates the outlier, which was excluded when fitting the overall index of switching costs. The *p*-values should be corrected for multiple comparisons (12 tests in total; all *p*-values exceed 0.548 after FDR correction), but we report the uncorrected values to show the most liberal estimate of the potential relationship between production and comprehension.

Additionally, we observed near-significant correlation in L1 mixing costs between modalities with *BF* close to 1 ($p = 0.061$, $BF_{01} = 0.90$). In contrast, we found weak evidence supporting the absence of correlations in switching costs across the three languages ($1 < BF_{01} < 2.5$), as well as in L2 mixing costs ($BF_{01} = 2.68$). After Benjamini-Hochberg False Discovery Rate (FDR) correction for 12 comparisons, the adjusted $ps_{FDR}$ were greater than 0.548.

Secondly, a radar chart was used to visualise all language control indices during production and comprehension (Figure 4(A)). Visual inspection suggests that (1) most indices exhibit greater magnitudes in production than in comprehension, and (2) the patterns of these indices show little similarity between the two modalities.

To statistically examine whether the magnitudes of these indices differ significantly between production and comprehension, a permutation test was performed on the differences in polygon area between modalities. The actual difference was calculated at the individual level by converting axis magnitudes into angles and Cartesian coordinates, computing the area using the *polyarea* function (Habel et al. 2025), subtracting comprehension from production, and then averaging the resulting values across participants. The null distribution was generated by randomly swapping half of the indices between modalities within each participant and computing the area difference using the same method as for the actual difference, repeated over 5000 permutations. The *p*-value was the proportion of permuted area differences greater than the actual difference. The result was statistically significant ($p < 0.001$; Figure 4(B)), confirming that the magnitudes of these language control indices in production were significantly greater than those in comprehension.

To statistically examine whether the pattern of these indices differs significantly between production and comprehension, a permutation test was performed on the cosine similarity between modalities. We first removed magnitude-related information by subtracting the minimum value within each type of index for each participant. The actual similarity was computed at the individual level by treating the indices as dimensions in a multidimensional space, representing production and comprehension as two vectors used to calculate cosine similarity, and then averaging the

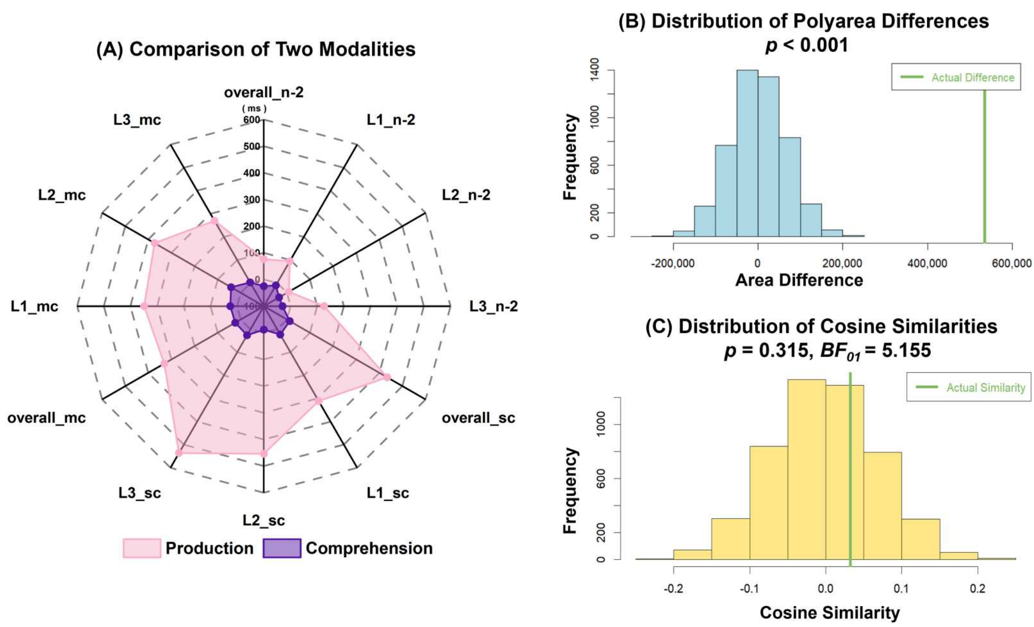


**Figure 4.** Radar chart representations of 12 indices for production and comprehension (A), along with permutation tests of polygon area (B) and cosine similarity (C). *n*-2, *n*-2 repetition costs; sc, switching costs; mc, mixing costs. Overall refers to the combined performance across all three languages.

resulting values across participants. This approach is commonly used to assess the similarity of behavioural patterns (Madsen et al. 2021; Roads and Love 2024). The null distribution was generated by randomly shuffling language labels within each index type and modality for each participant, and computing the cosine similarity using the same method as for the actual similarity. This process was repeated over 5000 permutations. The $p$-value was the proportion of permuted similarities greater than the actual similarity. The result was statistically non-significant ($p = 0.315$; Figure 4(C)), which does not permit a conclusive interpretation. Therefore, we additionally conducted a Bayesian paired-sample $t$-test using the *BayesFactor* package (Morey and Rouder 2024) to compare the actual similarity with the average similarity obtained from the 5000 permutations. The analysis yielded a Bayes factor ($BF_{01} = 5.155$), providing strong evidence in favour of the null hypothesis – that is, no shared pattern of language control indices between production and comprehension. Since both production and comprehension exhibited a similar pattern for mixing costs at the group level (L2 ≥ L1 > L3, Table 1), we further conducted cosine similarity and permutation tests specifically on mixing costs using the same approach. The results again indicated that the pattern similarity was no greater than random ($p = 0.161$, $BF_{01} = 3.401$), suggesting that no shared pattern existed between modalities at the individual level, even for mixing costs where both modalities showed L2 ≥ L1 > L3 at the group level.

Furthermore, we examined the number of participants who followed the L2 ≥ L1 > L3 pattern across three types of costs in both production and comprehension. For $n$-2 repetition costs, 2 participants exhibited this pattern in production and 4 in comprehension, with 1 participant overlapping between modalities; for switching costs, 2 in production and 3 in comprehension (no overlap); and for mixing costs, 16 in production and 8 in comprehension (only 4 overlapping).

## General discussion

This study investigated whether multilingual speech production and comprehension rely on shared or distinct language control mechanisms. We found that production exhibited strong aggregate effects on three types of indices (i.e. $n$-2 repetition, switching, and mixing costs), whereas comprehension only showed a significant group mean for mixing costs. Importantly, none of the 12 correlations reached significance, including near-significant trends that did not survive multiple comparison correction, and Bayesian analyses revealed substantial evidence for null effects in seven cases. Furthermore, permutation tests of polygon area revealed significant differences in the magnitude of language control indices between production and comprehension, whereas tests of cosine similarity showed no significant differences in the overall pattern of indices across modalities, as supported by Bayesian statistics. Thus, our findings provide compelling evidence that production and comprehension are governed by distinct language control mechanisms.

Firstly, our findings indicate that language control in production and comprehension shows little overlap, with production exhibiting larger control costs than comprehension. It is common to examine whether two tasks engage the same underlying mechanisms by correlations from an individual difference perspective in the literature (e.g. Ahn et al. 2020; Struys et al. 2019). We found no significant correlations between modalities across the three cost types, with most of the findings further supported by Bayesian statistics. Even the near-significant correlation in L1 mixing costs did not survive the FDR correction. Importantly, Bayesian statistics provided substantial evidence for the absence of correlations in seven out of the 12 cases. In addition, our radar chart and permutation tests of polygon area revealed that the magnitude of language control indices in production is significantly greater than in comprehension. The overall effects at the group level showed that $n$-2 repetition costs, switching costs, and mixing costs (331 ms) in production are all significant in production, where only mixing costs (24 ms) reached significance in comprehension of a much smaller magnitude. However, correlation analysis and polygon area calculation cannot determine whether the differences between production and comprehension reflect a qualitatively different mechanism or merely a weaker engagement of the same mechanism.

Secondly, our radar chart and permutation tests of cosine similarity revealed no shared pattern of language control indices between production and comprehension. This finding, also supported by Bayesian analyses, suggests that production and comprehension are governed by distinct language control mechanisms (e.g. Ahn et al. 2020; Li et al. 2024; Mosca and de Bot 2017), rather than a single set of mechanisms that are robust in production but weak in comprehension (de Bruin and Xu 2023; Declerck and Philipp 2017; Macizo, Bajo, and Paolieri 2012). Previous studies have mainly focused on switching costs and showed that performance patterns differed between language production and comprehension (Macizo, Bajo, and Paolieri 2012; Qi et al. 2009; Qi, Peng, and Ding 2010; Reynolds, Schlöffel, and Peressotti 2016). Similarly, neuroimaging studies have provided evidence supporting the dissociation of language control mechanisms in these modalities, based on activation patterns across multiple brain regions (Blanco-Elorrieta and Pylkkanen 2016, 2017; Giglio et al. 2022). In the present study, the absence of similarity in a comprehensive set of language control indices across modalities offers compelling support for the claim that production and comprehension rely on modality-specific control mechanisms during language switching.

Moreover, we sought to delineate the distinct language control mechanisms in production and comprehension, which might arise from distinct processing architectures. In production, language control is driven by internal intentions to articulation (endogenous; Grainger, Midgley, and Holcomb 2010), requiring both local inhibition of competing lexical candidates and global inhibition of the nontarget language schemas (Green 1998; Ma, Li, and Guo 2016), supported by the recruitment of inhibition-related regions like the dorsolateral prefrontal cortex (Blanco-Elorrieta and Pylkkanen 2016). In contrast, comprehension is driven by external input (exogenous; Grainger, Midgley, and Holcomb 2010) and focuses on accessing meaning. Cross-language activation may facilitate understanding (Liu and Chaouch-Orozco 2025), relying on conflict-monitoring regions like the anterior cingulate cortex to monitor activated lexical representations (Blanco-Elorrieta and Pylkkanen 2016). Unlike the global control exerted proactively before and during language production, global control in comprehension is likely recruited after word identification, and adapts flexibly to task demands (Dijkstra and van Heuven 2002).

Additionally, we have delved deeper into the costs by specific languages. The ICM posits that more dominant languages require greater inhibition. If so, all cost types should exhibit a pattern reflecting language dominance: most dominant (L2, Mandarin) > dominant (L1, Cantonese) > least dominant (L3, English). Given that both L1 and L2 are native languages for our participants, a pattern of L2 = L1 > L3 is also plausible. However, neither modality showed this expected dominance-based pattern in local control indices; but this pattern only emerged in global control in both modalities. This finding aligns with prior research that consistently report this pattern mainly in mixing costs, but rarely in switching (Gade et al. 2021; Goldrick and Gollan 2023) or *n*-2 repetition costs (Declerck and Koch 2022; Koch et al. 2024). We propose that dominant languages may already be over-inhibited at the global level, rendering them contextually nondominant during local control and potentially explaining the observed deviations from ICM predictions. Moreover, factors such as language proficiency, language usage, and task design may drive the global-level over-inhibition, which might contribute to inconsistent findings across studies. In the present study, we compared production and comprehension within the same participants from the perspective of individual differences, thereby minimising variability across studies and participants. Notably, although both modalities showed expected dominance patterns at the group level (but not at the local level), the permutation test of cosine similarity for mixing costs provided evidence of no significant similarity between modalities. Consistently, only four participants showed the predicted L2 ≥ L1 > L3 pattern in both modalities. These results suggest qualitative distinctions in language control mechanisms between production and comprehension at both the global and local levels.

It is worth noting that switching costs in production were substantially larger in our study (434 ms, averaged across three languages) than typically reported (30–120 ms, e.g. de Bruin and Xu 2023; Mosca and de Bot 2017; Qi, Peng, and Ding 2010), despite a high switch proportion

(81.46%) that should reduce switching costs (Duthoo et al. 2012; Geddert and Egner 2022; Siqi-Liu and Egner 2020). We reason that switching among three languages may add complexity, as trilinguals generally experience greater switching costs among three languages (200–230 ms; Gong, Hansen, and Chen 2024; Linck, Schwieter, and Sunderman 2011) than two languages (30–150 ms; Chen et al. 2023). However, the costs observed in our study still exceed these ranges. We also speculate that our participants' linguistic profiles may play a role, though further research is needed to clarify this. Notably, despite both modalities employing identical paradigms and trial sequences, this inflated switching cost was exclusive to the production task – supporting our broader conclusion that production and comprehension engage distinct language control mechanisms.

An unavoidable limitation of the present study was unequal trial distribution across Trial type. Nevertheless, it deepens our theoretical insight into the language control mechanisms between speech production and comprehension. The findings suggest that aligning instruction with the specific demands of each modality can effectively enhance the cognitive skills essential for speaking and listening.

## Conclusions

In this study, we recruited Cantonese–Mandarin–English trilinguals to perform both spoken word production and auditory word comprehension tasks, combining the $n$-2 repetition and language-switching paradigms within both single- and mixed-language contexts. Moving beyond prior studies that focused narrowly on group-level switching costs, we systematically assessed a broader set of language control indices – mixing, switching, and $n$-2 repetition costs – across modalities using individual-difference approaches. Results revealed that language control in production and comprehension differs not only in strength but also potentially in nature. That is, all indices were robust in production, but only mixing costs reached significance in comprehension, without any cross-modality correlations. In addition, radar plots with permutation tests revealed larger polygon areas for production than for comprehension, while the absence of similar index patterns between modalities was detected by permutation tests. These findings underscore the importance of distinguishing between modalities in language control models and highlight the value of multi-index, individual-differences approaches in bilingual and multilingual research.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

## Funding

This work was supported by the National Natural Science Foundation of China [grant number 32100854 to JW], the China Scholarship Council [grant number 202308120071 to JW], the National Social Science Fund of China [grant number 20BYY098 to MY], and the Australia Research Council [grant number DP 210102789 to XW].

## Data availability statement

Data and code can be found at OSF: https://osf.io/vxck6/?view_only=f3ceeab6ace64020ab73e925fdc8fd2d

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to polish the language. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

## ORCID

Miao Yu http://orcid.org/0000-0003-3031-8856

## References

Abutalebi, J., and D. Green. 2008. "Control Mechanisms in Bilingual Language Production: Neural Evidence from Language Switching Studies." *Language and Cognitive Processes* 23 (4): 557–582. https://doi.org/10.1080/01690960801920602.

Ahn, D., M. J. Abbott, K. Rayner, V. S. Ferreira, and T. H. Gollan. 2020. "Minimal Overlap in Language Control across Production and Comprehension: Evidence from Read-Aloud versus Eye-tracking Tasks." *Journal of Neurolinguistics* 54:100885. https://doi.org/10.1016/j.jneuroling.2019.100885.

Blanco-Elorrieta, E., and L. Pylkkanen. 2016. "Bilingual Language Control in Perception versus Action: Meg reveals Comprehension Control Mechanisms in Anterior Cingulate Cortex and Domain-General Control of Production in Dorsolateral Prefrontal Cortex." *The Journal of Neuroscience* 36 (2): 290–301. https://doi.org/10.1523/JNEUROSCI.2597-15.2016.

Blanco-Elorrieta, E., and L. Pylkkänen. 2017. "Bilingual Language Switching in the Laboratory versus in the Wild: The Spatiotemporal Dynamics of Adaptive Language Control." *The Journal of Neuroscience* 37 (37): 9022–9036. https://doi.org/10.1523/JNEUROSCI.0553-17.2017.

Braver, T. S. 2012. "The Variable Nature of Cognitive Control: A Dual Mechanisms Framework." *Trends in Cognitive Sciences* 16 (2): 106–113. https://doi.org/10.1016/j.tics.2011.12.010.

Chen, J., J. Wang, Z. Cairang, Y. Xiong, and C. Zhang. 2023. "Examining Non-task Language Effect on Production-Based Language Switching: Evidence from Tibetan-Chinese-English Trilinguals." *International Journal of Bilingualism* 27 (5): 795–814. https://doi.org/10.1177/13670069221123760.

Costa, A., M. Miozzo, and A. Caramazza. 1999. "Lexical Selection in Bilinguals: Do Words in the Bilingual's Two Lexicons Compete for Selection?" *Journal of Memory and Language* 41 (3): 365–397. https://doi.org/10.1006/jmla.1999.2651.

de Bruin, A., L. J. Hoversten, and C. D. Martin. 2023. "Interference between Non-native Languages during Trilingual Language Production." *Journal of Memory and Language* 128: 1–17. https://doi.org/10.1016/j.jml.2022.104386.

de Bruin, A., and T. Xu. 2023. "Language Switching in Different Contexts and Modalities: Response-Stimulus Interval Influences Cued-Naming but Not Voluntary-Naming or Comprehension Language-Switching Costs." *Bilingualism: Language and Cognition* 26 (2): 402–415. https://doi.org/10.1017/S1366728922000554.

Declerck, M., and J. Grainger. 2017. "Inducing Asymmetrical Switch Costs in Bilingual Language Comprehension by Language Practice." *Acta Psychologica* 178:100–106. https://doi.org/10.1016/j.actpsy.2017.06.002.

Declerck, M., and I. Koch. 2022. "The Concept of Inhibition in Bilingual Control." *Psychological Review* 130 (4): 953–976. https://doi.org/10.1037/rev0000367.

Declerck, M., I. Koch, J. A. Duñabeitia, J. Grainger, and D. N. Stephan. 2019. "What Absent Switch Costs and Mixing Costs during Bilingual Language Comprehension Can Tell us about Language Control." *Journal of Experimental Psychology: Human Perception and Performance* 45 (6): 771–789. https://doi.org/10.1037/xhp0000627.

Declerck, M., and A. M. Philipp. 2017. "Is Inhibition Implemented during Bilingual Production and Comprehension? *n*-2 Language Repetition Costs Unchained." *Language, Cognition and Neuroscience* 33 (5): 608–617. https://doi.org/10.1080/23273798.2017.1398828.

Dijkstra, T., and W. J. B. van Heuven. 1998. "The BIA Model and Bilingual Word Recognition." In *Localist Connectionist Approaches to Human Cognition*, edited by J. Grainger, and A. M. Jacobs, 189–225. Mahwah, NJ: Lawrence Erlbaum Associates.

Dijkstra, T., and W. J. B. van Heuven. 2002. "The Architecture of the Bilingual Word Recognition System: From Identification to Decision." *Bilingualism: Language and Cognition* 5 (3): 175–197. https://doi.org/10.1017/S1366728902003012.

Duthoo, W., W. De Baene, P. Wühr, and W. Notebaert. 2012. "When Predictions Take Control: The Effect of Task Predictions on Task Switching Performance." *Frontiers in Psychology* 3:282. https://doi.org/10.3389/fpsyg.2012.00282.

Eichert, N., D. Papp, R. B. Mars, and K. E. Watkins. 2020. "Mapping Human Laryngeal Motor Cortex during Vocalization." *Cerebral Cortex* 30 (12): 6254–6269. https://doi.org/10.1093/cercor/bhaa182.

Faul, F., E. Erdfelder, A. G. Lang, and A. Buchner. 2007. "G*Power 3: A Flexible Statistical Power Analysis Program for the Social, Behavioral, and Biomedical Sciences." *Behavior Research Methods* 39 (2): 175–191. https://doi.org/10.3758/bf03193146.

Gade, M., M. Declerck, A. M. Philipp, A. Rey-Mermet, and I. Koch. 2021. "Assessing the Evidence for Asymmetrical Switch Costs and Reversed Language Dominance Effects: A Meta-analysis." *Journal of Cognition* 4 (1): 55. https://doi.org/10.5334/joc.186.

Gambi, C., and R. J. Hartsuiker. 2016. "If You Stay, It Might Be Easier: Switch Costs from Comprehension to Production in a Joint Switching Task." *Journal of Experimental Psychology: Learning, Memory, and Cognition* 42 (4): 608–626. https://doi.org/10.1037/xlm0000190.

Geddert, R., and T. Egner. 2022. “No Need to Choose: Independent Regulation of Cognitive Stability and Flexibility Challenges the Stability-Flexibility Trade-off.” *Journal of Experimental Psychology: General* 151 (12): 3009. https://doi.org/10.1037/xge0001241.

Giglio, L., M. Ostarek, K. Weber, and P. Hagoort. 2022. “Commonalities and Asymmetries in the Neurobiological Infrastructure for Language Production and Comprehension.” *Cerebral Cortex* 32 (7): 1405–1418. https://doi.org/10.1093/cercor/bhab287.

Goldrick, M., and T. H. Gollan. 2023. “Inhibitory Control of the Dominant Language: Reversed Language Dominance Is the Tip of the Iceberg.” *Journal of Memory and Language* 130:104410. https://doi.org/10.1016/j.jml.2023.104410.

Gong, Y., K. Hansen, and J. Chen. 2024. “The (Dis)engagement of Different Components of Inhibitory Control in Trilingual Language Control..” *Journal of Experimental Psychology: Learning, Memory, and Cognition* 50 (7): 1095–1111. https://doi.org/10.1037/xlm0001315.

Grainger, J., K. Midgley, and P. J. Holcomb. 2010. “Chapter 14. Re-thinking the Bilingual Interactive-Activation Model from a Developmental Perspective (BIA-d).” In *Language Acquisition across Linguistic and Cognitive Systems*, edited by M. Kail and M. Hickmann, 267–284. New York, NY: John Benjamins. https://doi.org/10.1075/lald.52.18gra.

Green, D. W. 1998. “Mental Control of the Bilingual Lexico-Semantic System.” *Bilingualism: Language and Cognition* 1 (2): 67–81. https://doi.org/10.1017/S1366728998000133.

Habel, K., R. Grasman, R. Gramacy, P. Mozharovskyi, and D. Sterratt. 2025. *Geometry: Mesh generation and surface tessellation*. R package version 0.5.2. https://CRAN.R-project.org/package=geometry.

Hermans, D., T. Bongaerts, K. De Bot, and R. Schreuder. 1998. “Producing Words in a Foreign Language: Can Speakers Prevent Interference from Their First Language?” *Bilingualism: Language and Cognition* 1 (3): 213–229. https://doi.org/10.1017/S1366728998000364

Hirsch, P., M. Declerck, and I. Koch. 2015. “Exploring the Functional Locus of Language Switching: Evidence from a PRP Paradigm.” *Acta Psychologica* 161:1–6. https://doi.org/10.1016/j.actpsy.2015.07.010.

Jarosz, A. F., and J. Wiley. 2014. “What Are the Odds? A Practical Guide to Computing and Reporting Bayes Factors.” *Journal of Problem Solving* 7 (1): 2. https://doi.org/10.7771/1932-6246.1167.

JASP Team. 2018. JASP (Version 0.9) [Computer software].

Koch, I., M. Declerck, G. Petersen, D. Rister, W. Scharke, and A. M. Philipp. 2024. “Reassessing the Role of Language Dominance in *n*-2 Language Repetition Costs as a Marker of Inhibition in Multilingual Language Switching.” *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 50(9), 1516–1528. https://doi.org/10.1037/xlm0001333.

Li, C., and T. H. Gollan. 2022. “Language-switch Costs from Comprehension to Production Might Just Be Task-Switch Costs.” *Bilingualism: Language and Cognition* 25 (3): 459–470. https://doi.org/10.1017/s1366728921001061.

Li, C., K. J. Midgley, V. S. Ferreira, P. J. Holcomb, and T. H. Gollan. 2024. “Different Language Control Mechanisms in Comprehension and Production: Evidence from Paragraph Reading.” *Brain and Language* 248:105367. https://doi.org/10.1016/j.bandl.2023.105367.

Liang, B., Y. Li, W. Zhao, and Y. Du. 2023. “Bilateral Human Laryngeal Motor Cortex in Perceptual Decision of Lexical Tone and Voicing of Consonant.” *Nature Communications* 14 (1): 4710. https://doi.org/10.1038/s41467-023-40445-0.

Lim, J. H., and H. Ahn. 2024. “Trilingual Word Recognition: Phonological Priming Effects in Trilingual with Different-Script Languages.” *Korea Journal of English Language and Linguistics* 24:921–933. https://doi.org/10.15738/kjell.24..202409.921.

Linck, J. A., J. W. Schwieter, and G. Sunderman. 2011. “Inhibitory Control Predicts Language Switching Performance in Trilingual Speech Production.” *Bilingualism: Language and Cognition* 15 (3): 651–662. https://doi.org/10.1017/S136672891100054X.

Liu, H., and A. Chaouch-Orozco. 2025. “Language Control in Auditory Bilingual Comprehension: Uncovering Novel Evidence from the *n*-2 Repetition Paradigm.” *Language and Cognition* 17:e15. https://doi.org/10.1017/langcog.2024.43.

Ma, F., S. Li, and T. Guo. 2016. “Reactive and Proactive Control in Bilingual Word Production: An Investigation of Influential Factors.” *Journal of Memory and Language* 86:35–59. https://doi.org/10.1016/j.jml.2015.08.004.

Macizo, P., T. Bajo, and D. Paolieri. 2012. “Language Switching and Language Competition.” *Second Language Research* 28 (2): 131–149. https://doi.org/10.1177/0267658311434893.

Madsen, J., S. U. Júlio, P. J. Gucik, R. Steinberg, and L. C. Parra. 2021. “Synchronized eye Movements Predict Test Scores in Online Video Education.” *Proceedings of the National Academy of Sciences of the United States of America* 118 (5): e2016980118. https://doi.org/10.1073/pnas.2016980118.

Meuter, R. F. I., and A. Allport. 1999. “Bilingual Language Switching in Naming: Asymmetrical Costs of Language Selection.” *Journal of Memory and Language* 40 (1): 25–40. https://doi.org/10.1006/jmla.1998.2602.

Miller, J., and W. Schwarz. 2018. “Implications of Individual Differences in On-average Null Effects.” *Journal of Experimental Psychology: General* 147 (3): 377–397. https://doi.org/10.1037/xge0000367.

Morey, R., and J. Rouder. 2024. *BayesFactor: Computation of Bayes factors for common designs*. R package version 0.9.12-4.7.
Mosca, M., and K. de Bot. 2017. "Bilingual Language Switching: Production vs. Recognition." *Frontiers in Psychology* 8:934. https://doi.org/10.3389/fpsyg.2017.00934.
Peeters, D., E. Runnqvist, D. Bertrand, and J. Grainger. 2014. "Asymmetrical Switch Costs in Bilingual Language Production Induced by Reading Words." *Journal of Experimental Psychology: Learning, Memory, and Cognition* 40 (1): 284–292. https://doi.org/10.1037/a0034060.
Philipp, A. M., M. Gade, and I. Koch. 2007. "Inhibitory Processes in Language Switching: Evidence from Switching Language-Defined Response Sets." *European Journal of Cognitive Psychology* 19 (3): 395–416. https://doi.org/10.1080/09541440600758812.
Philipp, A. M., and I. Koch. 2009. "Inhibition in Language Switching: What Is Inhibited When Switching between Languages in Naming Tasks?" *Journal of Experimental Psychology: Learning, Memory, and Cognition* 35 (5): 1187–1195. https://doi.org/10.1037/a0016376.
Qi, Z., D. Peng, and G. Ding. 2010. "A Research on the Language Switching Process of Chinese-English Bilinguals under Unpredictable Conditions (in Chinese)." *Psychological Science* 33 (5): 1051–1053. https://doi.org/10.16719/j.cnki.1671-6981.2010.05.008.
Qi, Z., D. Peng, X. Xu, and H. Liu. 2009. "A Research on the Switching Process of Chinese-English Bilinguals in Language Production and Comprehension (in Chinese)." *Psychological Science* 32 (2): 356–359. https://doi.org/10.16719/j.cnki.1671-6981.2009.02.021.
Reynolds, M. G., S. Schlöffel, and F. Peressotti. 2016. "Asymmetric Switch Costs in Numeral Naming and Number Word Reading: Implications for Models of Bilingual Language Production." *Frontiers in Psychology* 6:2011. https://doi.org/10.3389/fpsyg.2015.02011.
Roads, B. D., and B. C. Love. 2024. "Modeling Similarity and Psychological Space." *Annual Review of Psychology* 75 (1): 215–240. https://doi.org/10.1146/annurev-psych-040323-115131.
Schwieter, J. W., and G. Sunderman. 2011. "Inhibitory Control Processes and Lexical Access in Trilingual Speech Production." *Linguistic Approaches to Bilingualism* 1 (4): 391–412. https://doi.org/10.1075/lab.1.4.02sch.
Siqi-Liu, A., and T. Egner. 2020. "Contextual Adaptation of Cognitive Flexibility Is Driven by Task- and Item-Level Learning." *Cognitive, Affective, and Behavioral Neuroscience* 20 (4): 757–782. https://doi.org/10.3758/s13415-020-00801-9.
Snodgrass, J. G., and M. Vanderwart. 1980. "A Standardized Set of 260 Pictures: Norms for Name Agreement, Image Agreement, Familiarity, and Visual Complexity." *Journal of Experimental Psychology: Human Learning and Memory* 6 (2): 174–215. https://doi.org/10.1037/0278-7393.6.2.174.
Struys, E., J. Surmont, P. Van de Craen, O. Kepinska, and M. Van den Noort. 2019. "Bilingual Language Control across Modalities: The Relationship between Mixed-Language Comprehension and Production." *Linguistic Approaches to Bilingualism* 9 (4-5): 542–568. https://doi.org/10.1075/lab.17020.str.
Thierry, G., and Y. J. Wu. 2007. "Brain Potentials Reveal Unconscious Translation during Foreign-Language Comprehension." *Proceedings of the National Academy of Sciences* 104 (30): 12530–12535. https://doi.org/10.1073/pnas.0609927104
Thomas, M. S. C., and A. Allport. 2000. "Language Switching Costs in Bilingual Visual Word Recognition." *Journal of Memory and Language* 43 (1): 44–66. https://doi.org/10.1006/jmla.1999.2700.
Wang, Q., X. Wu, Y. Ji, G. Yan, and J. Wu. 2022. "Second Language Proficiency Modulates the Dependency of Bilingual Language Control on Domain-General Cognitive Control." *Frontiers in Psychology* 13:810573. https://doi.org/10.3389/fpsyg.2022.810573.
Wu, J., Y. Ji, H. Qu, S. Zuo, J. Liang, J. Su, Q. Wang, G. Yan, and G. Ding. 2025. "Transcranial Magnetic Stimulation of the Right Inferior Frontal Gyrus Impairs Bilinguals' Performance in Language-Switching Tasks." *Cognition* 254:105963. https://doi.org/10.1016/j.cognition.2024.105963.
Wu, J., J. Yang, M. Chen, S. Li, Z. Zhang, C. Kang, G. Ding, and T. Guo. 2019. "Brain Network Reconfiguration for Language and Domain-General Cognitive Control in Bilinguals." *NeuroImage* 199:454–465. https://doi.org/10.1016/j.neuroimage.2019.06.022.
Wu, J., Z. Zhang, M. Chen, Q. Yuan, M. Zhang, J. Yang, C. Lu, and T. Guo. 2020. "Language Context Tunes Brain Network for Language Control in Bilingual Language Production." *Neuropsychologia* 147:107592. https://doi.org/10.1016/j.neuropsychologia.2020.107592.
Wu, J., H. Zhao, X. Wu, Q. Liu, J. Su, Y. Ji, and Q. Wang. 2025. "Word Concreteness Modulates Bilingual Language Control during Reading Comprehension." *Journal of Experimental Psychology. Learning, Memory, and Cognition* 51 (3): 425–434. https://doi.org/10.1037/xlm0001385.
Yang, T., Z. G. Cai, W. Lin, and R. Wang. 2024. "Modality-general and Modality-Specific Bilingual Control Mechanisms in Spoken and Written Productions." *Bilingualism: Language and Cognition* 27 (5): 862–873. https://doi.org/10.1017/S1366728924000166.